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DSCI-D 532

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Final Project: Analyzing League of Legends eSports Data

Project Overview

There is a lack of centralized, accessible, and user-friendly data for professional League of Legends (LoL) eSports. Although Riot Games maintains its own internal database for employees and industry partners, that information is not publicly available. Riot also places strict limitations on the public use of its graphics, branding, and certain eSport assets. As a result, fans, analysts, coaches and players often rely on third-party websites that vary in completeness, accuracy, and ease of access.

This application aims to provide a free, open-source, and user-friendly database that centralizes the most important statistics for professional and developmental LoL competition.

GitHub Repo

https://github.com/jhurley00/finalproject_532/tree/main

Deployed Web Application

https://lol-esports-database.onrender.com/

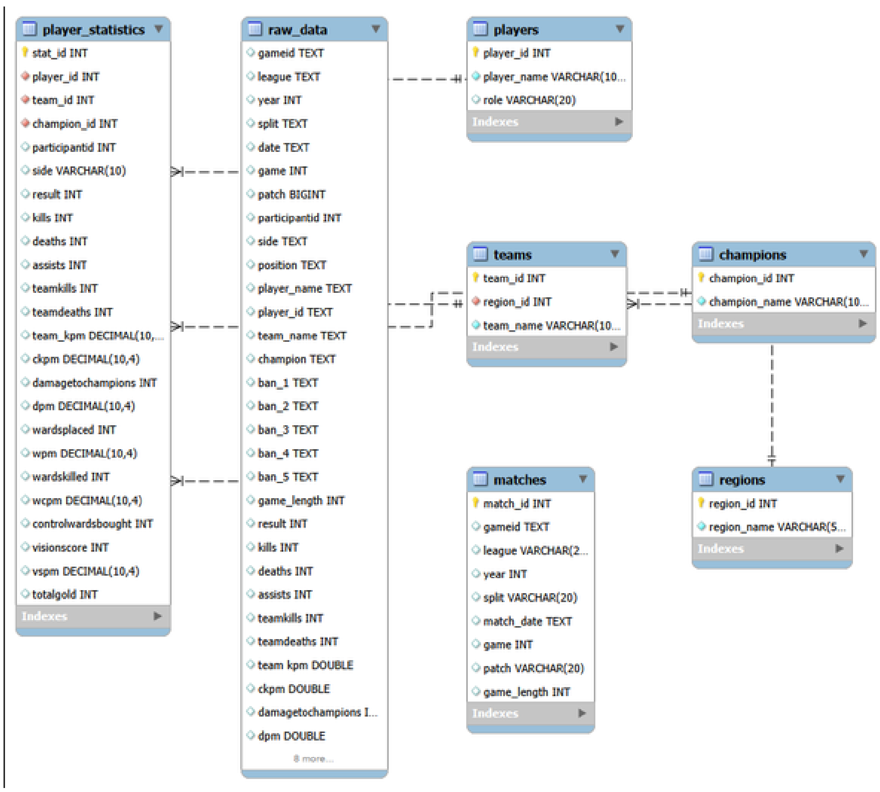
Database Development

The dataset used in this project was obtained from Oracle’s Elixir, a community-maintained project that provides professional LoL eSports statistics (Sevenhuysen, n.d.). The original data was distributed as a comma-separated values file, where each row represents a single player’s performance in a professional match during the 2023-2024 competitive seasons. In addition to the player's records, each match contains two additional rows, summarizing the overall performance of each team.

During the development of the database, significant data preprocessing and SQL refinement were required to create a reliable dataset. The original Oracle's Elixir CSV contained over 133,000 rows and many unnecessary or incomplete columns, which caused import and performance issues. The dataset was reduced to approximately 26,000 rows by removing redundant fields and retaining only the statistics needed for analysis. The database schema was also simplified by normalizing the data into separate tables for regions, teams, players, champions, matches, and player statistics.

SQL scripts were developed to populate these tables using INSERT IGNORE statements and appropriate foreign key relationships. Several queries were revised throughout development to include aggregating player statistics, calculating team win rates, identifying the most-played champions, and displaying player and team information within the Flask application.

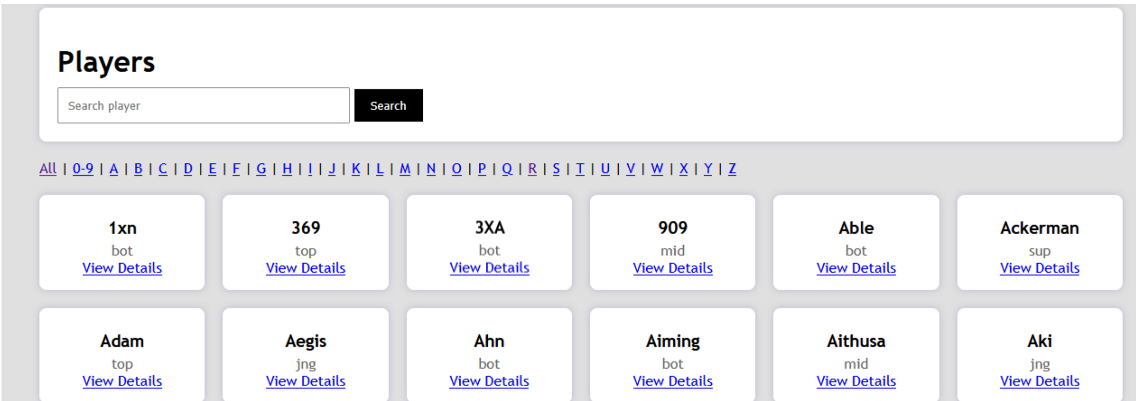
Final Database Design



Final Application

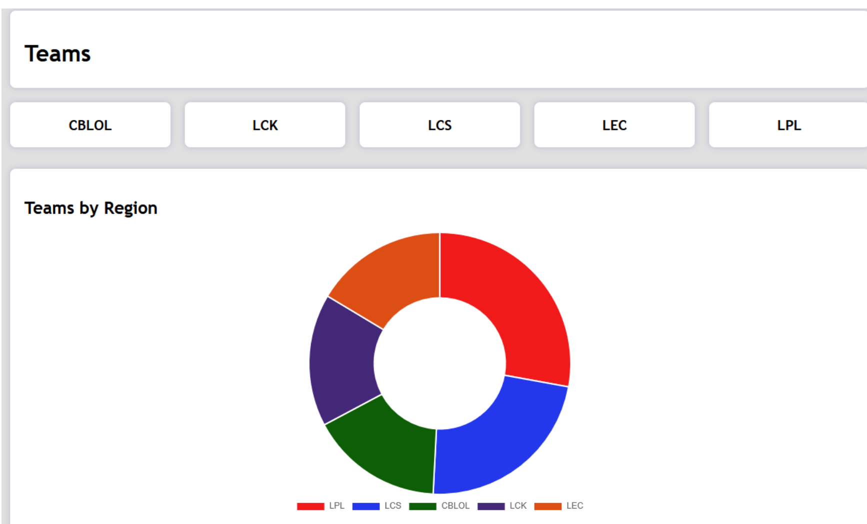
For this project, I included a few simple features such as searching, filtering, and interactable UI elements. Each page has an interactive feature.

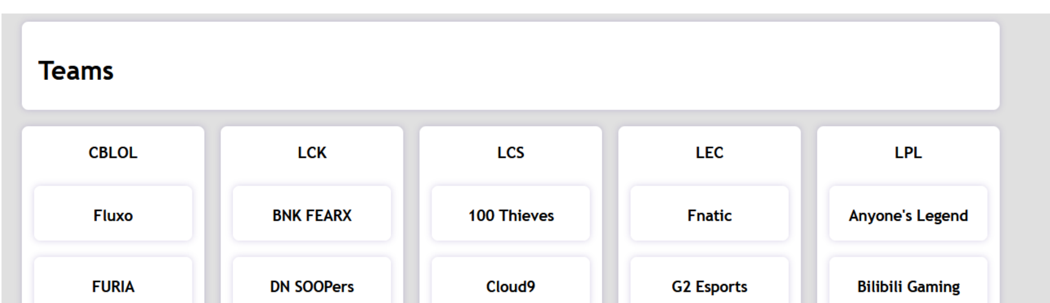
Player Search



User can view player statistics for each player and can search for players by both search bar and clicking on the first initial of their in-game name.

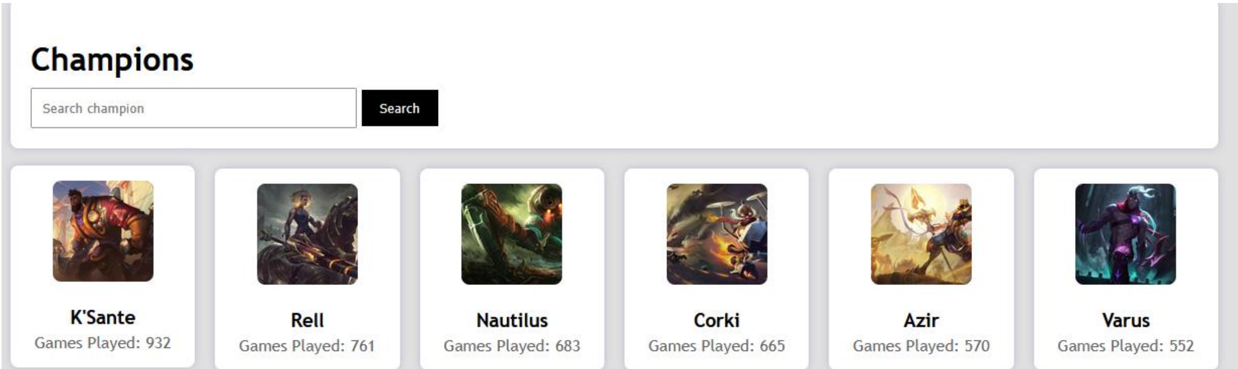
Teams





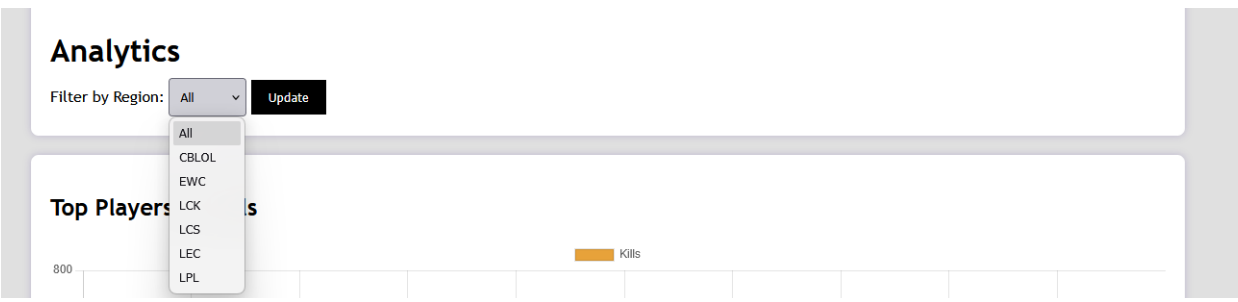
Users can view teams by individual region, see the distribution of professional teams by region, and hover to see number of teams per region.

Champions



Users can search by champion name and see the name, in-game splash art, and how frequently they were played during the season.

Analytics



Users can filter between regional and all stats for top players by kills, top champions by games played, and top teams by win rate. Teams that have >=50% win-rate are green, whereas teams with <= 50% win-rate are red.

Technical Summary

Technologies Used

- MySQL – Database management system, design, and SQL development
- Python (Flask) -- Backend framework used to connect the web application to MySQL database
- HTML & CSS – User interface and overall visual design
- JavaScript – Chart.js for analytics and dashboards
- Microsoft Excel – EDA, cleaning data, preprocessing
- Render – Hosts the web application
- Aiven – Hosts the database

Team Member Contributions

I completed this project by myself from start to finish. In total, I spent approximately 45 hours on this project. A significant portion of the time was spent learning Flask and how to integrate it with MySQL. Additional time was spent cleaning the dataset and debugging the web interface.

Reflection

The biggest challenge I faced in this project was cleaning and importing the original dataset. I spent most of the time trying to figure out how much of the data I was using. I also ran into issues using MySQL workbench on my laptop that currently runs Linux. It eventually required me to restart the project on my Windows desktop, which significantly limited how much time I could actually spend on the project since I am not home often. It ultimately made the project much more stable as MySQL Workbench is not supported on Linux. Also, learning Flask posed a significant challenge. Since it was my first time using it, there was a steep learning curve in understanding what was going on.

The most enjoyable part of the project was designing the web application! I have completed a few web-design projects that use HTML, CSS, and Javascript, so it was exciting to build on those skills with a framework that is used in the industry!

The first improvement I would make would be to expand the filtering and search options. I would also include more of the original data set. In this project, I limited the data used to Tier 1, which is comparable to the highest level of competition in traditional sports (NFL, NBA, etc.). In the future, I would like to include Tier 2, challenger, and collegiate teams, as these players eventually promote to Tier 1.

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